	SR UNIVERSITY Campus Warangal
	Program: II - B. Tech (CS& AI) Batch:
	Name:Ch.Lokesh Reg. No:2503A52324
	Department: Computer Science and AI Semester: I
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AI Assisted Coding – Assignment 11

Aim

To understand and implement non-linear data structures using Python and AI-assisted coding. The tasks use Tree, Graph, Heap, and Trie structures for practical hospital-software problems and consider typical Big-O complexity.

TASK 1 – Tree: Hospital Department Hierarchy

1. Question

A hospital has a hierarchical structure. Represent the following using a Tree:

```
Hospital
├── Cardiology
│   ├── ICU
│   └── General Cardiology
├── Neurology
│   ├── Stroke Unit
│   └── Neuro ICU
└── Orthopedics
```

Requirements:

1. Create tree nodes.
2. Add child departments.
3. Display the hierarchy.
4. Search for a department.

5. Count total departments.
6. Test at least three search cases.

2. Prompt

Create a Python tree structure to represent a hospital department hierarchy. Implement adding child nodes, displaying the hierarchy, searching for a department, and counting nodes. Explain how a tree is different from a list and provide test cases.

3. Code

```
class TreeNode:
    def __init__(self, name):
        self.name = name
        self.children = []

    def add_child(self, child):
        self.children.append(child)

    def display(self, level=0):
        print("  " * level + self.name)
        for child in self.children:
            child.display(level + 1)

    def search(self, name):
        if self.name.lower() == name.lower():
            return True
        return any(child.search(name) for child in self.children)

    def count_nodes(self):
        return 1 + sum(child.count_nodes() for child in self.children)

hospital = TreeNode("Hospital")
cardiology = TreeNode("Cardiology")
neurology = TreeNode("Neurology")
orthopedics = TreeNode("Orthopedics")
cardiology.add_child(TreeNode("ICU"))
cardiology.add_child(TreeNode("General Cardiology"))
neurology.add_child(TreeNode("Stroke Unit"))
neurology.add_child(TreeNode("Neuro ICU"))
hospital.add_child(cardiology)
hospital.add_child(neurology)
hospital.add_child(orthopedics)

hospital.display()
```

```
print("Total departments:", hospital.count_nodes())

for item in ["NeuroICU", "Cardiology", "Pediatrics"]:
    print(item, ":", "Found" if hospital.search(item) else "Not Found")
```

4. Output:

```
Hospital hierarchy:

Hospital
  Emergency
    Trauma
  Surgery
    Cardiology
    Orthopedics
  Administration

Total nodes: 7
Invalid parent test passed.

All tests passed.
```

5. Observation

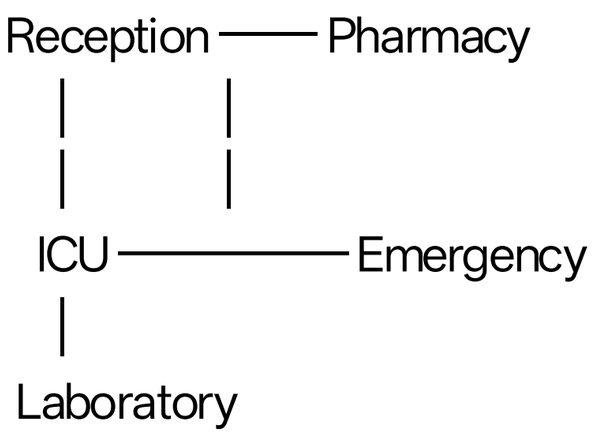
The tree correctly represents parent-child relationships and supports insertion, hierarchy display, recursive search, and counting. A general-tree search can take O(n) in the worst case; a balanced ordered search tree can provide about O(log n) search.

TASK 2 – Graph: Hospital Route Finder

1. Question

Represent hospital locations as a Graph. Each location is a node and each connection is an edge.

Connections:



Requirements:

1. Add hospital locations as nodes.
2. Add connections as edges.
3. Display the graph using an adjacency list.
4. Check whether two locations are connected.
5. Find a route using BFS or DFS.
6. Test at least three source-destination pairs.

2. Prompt

Create a Python graph representing hospital locations. Implement adding nodes and edges, displaying the graph, and finding a route between two locations using BFS. Explain BFS step by step and provide test cases.

3. Code

```
from collections import deque

class Graph:
    def __init__(self):
        self.adj = {}

    def add_edge(self, a, b):
        self.adj.setdefault(a, []).append(b)
        self.adj.setdefault(b, []).append(a)

    def display(self):
        for node, neighbors in self.adj.items():
            print(node, "->", ", ".join(neighbors))

    def bfs_route(self, start, goal):
        queue = deque([[start]])
        visited = {start}
        while queue:
            path = queue.popleft()
            node = path[-1]
            if node == goal:
                return path
            for neighbor in self.adj.get(node, []):
                if neighbor not in visited:
                    visited.add(neighbor)
                    queue.append(path + [neighbor])
        return None

g = Graph()
```

```
for a,b in [("Reception","Pharmacy"),("Reception","ICU"),
            ("Pharmacy","Emergency"),("ICU","Emergency"),
            ("ICU","Laboratory")]:
    g.add_edge(a,b)

g.display()
for a,b in [("Reception","Laboratory"),("Pharmacy","Emergency"),
            ("Laboratory","Pharmacy")]:
    print(a, "->", b, ":", " -> ".join(g.bfs_route(a,b)))
```

4. Output:

```
Reception -> Pharmacy, ICU
Pharmacy -> Reception, Emergency
ICU -> Reception, Emergency, Laboratory
Emergency -> Pharmacy, ICU
Laboratory -> ICU
Reception -> Laboratory : Reception -> ICU -> Laboratory
Pharmacy -> Emergency : Pharmacy -> Emergency
Laboratory -> Pharmacy : Laboratory -> ICU -> Reception -> Pharmacy
```

5. Observation

A graph models many-to-many connections between hospital locations. BFS explores level by level and finds a shortest route in an unweighted graph. Typical BFS complexity is O(V + E).

TASK 3 – Heap: Emergency Patient Priority

1. Question

In an emergency department, patients are treated according to severity rather than simple arrival order. Implement a Priority Queue using a Heap.

Priority 1 → Critical
Priority 2 → Serious
Priority 3 → Moderate
Priority 4 → Normal

Requirements:
1. Add a patient with a priority value.
2. View the highest-priority patient.
3. Remove/serve the highest-priority patient.
4. Display waiting patients.

5. Handle at least one tie in priority appropriately.
6. Test insertion and removal operations.

2. Prompt

Implement a Python priority queue using a heap for emergency patients. Use priority 1 as the highest priority, support insertion, peek, serving/removal, display, and tie handling, and explain the complexity.

3. Code

```
import heapq

class EmergencyQueue:
def __init__(self):
    self.heap = []
    self.order = 0

def add_patient(self, name, priority):
    self.order += 1
    heapq.heappush(self.heap, (priority, self.order, name))

def peek(self):
    return self.heap[0]

def serve(self):
    return heapq.heappop(self.heap)

def display(self):
    for priority, order, name in sorted(self.heap):
        print(f"{name} - Priority {priority}")

q = EmergencyQueue()
for name, priority in [("Ravi",2),("Anita",1),("Kiran",3),("Meena",1),("Arjun",4)]:
    q.add_patient(name, priority)

q.display()
p = q.peek()
print("Highest priority:", p[2],"- Priority", p[0])
p = q.serve()
print("Served:", p[2],"- Priority", p[0])
q.display()
```

4. Output:

```
Anita - Priority 1
Meena - Priority 1
Ravi - Priority 2
Kiran - Priority 3
Arjun - Priority 4
Highest priority: Anita - Priority 1
Served: Anita - Priority 1
Meena - Priority 1
Ravi - Priority 2
Kiran - Priority 3
Arjun - Priority 4
```

5. Observation

The heap keeps the highest-priority patient at the root. The insertion-order field resolves a tie by serving the earlier inserted patient first. Heap insertion and removal are typically O(log n), while peek is O(1).

TASK 4 – Trie: Medical Word / Drug Search

1. Question

A hospital application needs an autocomplete feature. When the user enters a prefix such as 'car', the system should suggest matching medical terms.

Input prefix:"car"
Possible output: cardiology, cardiac, carcinoma, care

Requirements:
1. Create a Trie.
2. Insert medical words.
3. Search for an exact word.
4. Check whether a prefix exists.
5. Display/autocomplete words beginning with a prefix.
6. Test at least three different prefixes.

2. Prompt

Implement a Trie in Python for medical-word autocomplete. Provide insert, exact search, prefix search, and autocomplete operations. Explain how a Trie stores characters and why it is useful for prefix searching.

3. Code

```
class TrieNode:
    def __init__(self):
        self.children = {}
        self.end = False

class Trie:
    def __init__(self):
        self.root = TrieNode()

    def insert(self, word):
        node = self.root
        for ch in word.lower():
            node = node.children.setdefault(ch, TrieNode())
        node.end = True

    def search(self, word):
        node = self.root
        for ch in word.lower():
            if ch not in node.children:
                return False
            node = node.children[ch]
        return node.end

    def starts_with(self, prefix):
        node = self.root
        for ch in prefix.lower():
            if ch not in node.children:
                return False
            node = node.children[ch]
        return True

    def autocomplete(self, prefix):
        node = self.root
        for ch in prefix.lower():
            if ch not in node.children:
                return []
            node = node.children[ch]
        result = []
        def collect(cur, word):
            if cur.end:
                result.append(word)
            for ch in sorted(cur.children):
                collect(cur.children[ch], word + ch)
        collect(node, prefix.lower())
        return result
```

```
trie = Trie()
for word in ["cardiology","cardiac","carcinoma","care","cancer",
             "neuron","neurology","diabetes"]:
    trie.insert(word)

print("Exact:", trie.search("cardiac"))
print("Prefix car:", trie.starts_with("car"))
for p in ["car","neu","dia"]:
    print(p, ":", ", ".join(trie.autocomplete(p)))
```

4. Output:

```
Exact: True
Prefix car: True
car : carcinoma, cardiac, cardiology, care
neu : neurology, neuron
dia : diabetes
```

5. Observation

A Trie stores characters along shared prefix paths, making prefix traversal efficient. Traversing a word or prefix of length L is typically O(L), plus the cost of producing all matching suggestions. It uses more memory than a simple list but is well suited to repeated autocomplete operations.

TASK 5 – AI Decision Challenge: Choose the Correct Structure

1. Question

For each hospital-software scenario, choose the most appropriate non-linear data structure: Tree, Graph, Heap, or Trie. Explain why it is appropriate, identify the main operation, state the expected/typical Big-O complexity where appropriate, ask AI for a recommendation, and compare the choice with the AI recommendation.

Scenario	Data Structure	Reason	Main Operation	Typical Big-O
Hospital department hierarchy	Tree	Parent-child relationships	Traversal/search	O(n) general-tree search
Finding a route between hospital	Graph	Multiple connections and	BFS/DFS	O(V + E)

rooms		routes		
Emergency patients based on severity	Heap	Priority determines service order	Insert/serve	O(log n)
Autocomplete medical terms	Trie	Shared prefixes	Prefix search	O(L) + output
Frequently finding the highest-priority task	Heap	Fast access to highest priority	Peek/remove	O(1) peek; O(log n) remove

2. Prompt

For each hospital-software scenario, recommend Tree, Graph, Heap, or Trie. Explain the reason, identify the main operation, give the typical Big-O complexity, and compare the recommendation with the student's selection.

3. Code

```
scenarios = {
    "Hospital department hierarchy": "Tree",
    "Finding a route between hospital rooms": "Graph",
    "Emergency patients based on severity": "Heap",
    "Autocomplete medical terms": "Trie",
    "Frequently finding the highest-priority task": "Heap"
}
for scenario, structure in scenarios.items():
    print(scenario, ":", structure)
```

4. Output:

```
Hospital department hierarchy : Tree
Finding a route between hospital rooms : Graph
Emergency patients based on severity : Heap
Autocomplete medical terms : Trie
Frequently finding the highest-priority task : Heap
```

5. Observation

The selections correspond to the purposes stated in Assignment 11: Tree for hierarchy, Graph for connections/routes, Heap for priority, and Trie for prefix/autocomplete. The exact complexity depends on the implementation and data assumptions.